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Executive Summary:

Cyclistic is a bike-share company looking to increase the number of annual memberships by converting casual riders into members. This analysis examines how casual riders and annual members use Cyclistic's bike-share service and identifies differences in their riding patterns that may help inform membership marketing strategies.

Business Task:

How do annual members and casual riders use Cyclistic bikes differently?
Find the difference between casual riders and annual memberships in the Cyclistic program. The marketing agency is trying to target casual riders and transform them into annual memberships since they are already informed of the Cyclistic genre.

Stakeholders:

Primary Stakeholder: Lily Moreno, The director of Marketing.

Ask:

How do annual members and casual riders use Cyclistic bikes differently?
Why would casual riders buy Cyclistic annual membership?
How can Cyclistic use digital media to influence casual riders to become members?

Prepare:

Dataset: 12 Raw Data zip files
Source: Motivate International Inc., License: [Data License Agreement | Divvy Bikes](#)
Time Period: April 2020- April 2021
Important Columns: member_casual, day_of_week

Process:

During the initial review of the monthly Cyclistic datasets, I identified missing values in several end-station and geographical fields, including the end station name, and the end station ID, end latitude, and end longitude. Rather than immediately removing these records, I will quantify the missing values and determine whether the records remain useful for analysis. This will help prevent unnecessary loss of valid trip data. After I began to combine the twelve months into one table for further analysis, I noticed a data-quality/type issue and handled it deliberately. After combining all twelve months together I searched for the difference of total rides between casual riders and members. The average duration of casual and member riders. Then compared the days out of the week.

Tools used:

BigQuery
SQL
Excel/Google Sheets
Tableau

SQL Findings and Analysis:

SQL was used to analyze Cyclistic ride data and identify differences in usage patterns between casual riders and annual members. The analysis focused on ride volume by rider type, day of the week, and hour of the day.

Ride Volume by Rider Type

The first query compared the total number of rides and average ride duration between casual riders and annual members. This provided an overall view of how frequently each rider type uses the Cyclistic service.

The results showed differences in overall riding behavior between the two groups, providing a starting point for further analysis of when each type of rider is most active.

Ride Volume by Day of the Week

The second analysis examined the number of rides taken by casual riders and annual members on each day of the week.

A notable difference was that casual riders had stronger ride activity on weekends compared with weekdays. Saturday had the highest ride volume among casual riders. This pattern suggests that casual riders may be more likely to use Cyclistic for recreational or leisure activities.

Annual members showed a more consistent pattern of usage throughout the week, indicating that their use of the service may be more closely connected to regular routines.

Ride Volume by Hour

The third analysis examined ride volume by hour to identify the times when casual riders and annual members were most active.

Comparing hourly ride volume provides additional insight into how the two rider groups use the service throughout the day. These patterns can help Cyclistic identify the periods when casual riders are most engaged and potentially provide opportunities for targeted membership marketing.

Overall Finding

The SQL analysis identified meaningful differences in how casual riders and annual members use Cyclistic. Casual riders showed stronger weekend activity, particularly on Saturday, while annual members demonstrated more consistent usage throughout the week. These differences provide a foundation for developing targeted strategies to encourage casual riders to become annual members.

Recommendations:

Based on the differences observed between casual riders and annual members, Cyclistic should focus on converting frequent casual riders into annual members through targeted marketing strategies.

1. **Target casual riders during high-usage weekends.**
Saturday has the highest ride volume among casual riders, making weekends an important opportunity to promote annual memberships. Cyclistic could display membership promotions through the app, website, or digital marketing channels during periods when casual rider activity is highest.
2. **Create messaging focused on the value of membership.**
Marketing should communicate the benefits of becoming an annual member and show casual riders how membership could provide greater value for riders who use Cyclistic frequently. Messaging could emphasize convenience, savings, and the ability to use the service throughout the year.
3. **Develop promotions designed to encourage conversion.**
Cyclistic could test limited-time membership offers aimed at frequent casual riders, such as

introductory membership discounts or incentives for upgrading after multiple rides. These promotions could be especially effective when presented during periods of high casual-rider activity.

4. **Continue monitoring rider behavior.**

Cyclistic should continue analyzing rider activity by rider type, day, and time to evaluate whether targeted marketing efforts increase annual membership conversions. Future analysis could also examine ride duration, location, and seasonal patterns to further refine marketing strategies.

10. Conclusion:

This analysis examines differences in bike-share usage between Cyclistic's casual riders and annual members, with the goal of identifying opportunities to convert casual riders into annual memberships.

The analysis found a clear difference in riding patterns between the two groups. Casual riders showed stronger usage on weekends compared with weekdays, with **Saturday representing the highest ride volume for casual riders and surpassing their weekday ride activity**. This suggests that casual riders may be more likely to use Cyclistic for recreational or leisure purposes, particularly during the weekend.

In comparison, annual members demonstrated a more consistent pattern of usage throughout the week. These differences in riding behavior provide an opportunity for Cyclistic to develop targeted membership marketing strategies aimed at casual riders who are already actively using the service.

Based on these findings, Cyclistic should consider targeting casual riders with membership promotions and messaging during high-usage weekend periods. Converting frequent weekend riders into annual members could help increase membership while building on riding behaviors that are already established among casual customers.